VoxNeuro: Accessible and Leakage-Safe Parkinson's Screening Support via Multi-View Grassmannian Speech Analysis

Laiba Khan[1,*] and Manas Waghe[2]

[1] Lexington High School, Lexington, MA 02421, USA; 27stu469@lexingtonma.org
[2] Worcester Polytechnic Institute, Worcester, MA 01609, USA; mywaghe@wpi.edu
* Correspondence: 27stu469@lexingtonma.org

Abstract: Parkinson's disease (PD) can alter phonation and articulation, making repeated speech a candidate low-burden digital biomarker. However, public speech datasets typically contain several recordings per participant. Record-wise cross-validation can therefore place the same individual in both training and test sets, so reported scores measure recognition of an observed person rather than generalization to an unseen participant. VoxNeuro treats the subject, not the recording, as the statistical observation. Each subject's three recordings, standardized within the outer training fold, define a rank-three Grassmann subspace and a Euclidean mean–variability vector. Gaussian kernels on the two views are fused into a positive-semidefinite kernel for support vector classification. For imbalanced data, minority subjects are synthesized along Grassmann geodesics with matched Euclidean interpolation. Under five-fold subject-disjoint evaluation, the fused model attained the highest mean balanced accuracy and macro-F1 among the audited comparators: 0.850 and 0.847 on UCI-489 and, on PD-252, 0.739 ± 0.012 and 0.752 ± 0.013 (mean $\pm$ SD across 24 augmentation seeds); margins over a matched Euclidean ablation were positive in 23/24 (macro-F1) and 16/24 (balanced-accuracy) draws. Permutation analysis identified the tunable Q-factor wavelet transform block as the dominant Euclidean-summary contributor. These results support subject-disjoint, geometry-aware evaluation as a default for repeated-measure speech biomarkers.

Keywords: Parkinson's disease; speech biomarker; Grassmann manifold; kernel fusion; imbalanced learning; subject-disjoint cross-validation; support vector machine; digital health

1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder involving nigrostriatal dopaminergic loss, basal ganglia circuit dysfunction, and heterogeneous motor and non-motor manifestations [1–3]. Its diagnosis remains clinical [4,5], and its global burden has grown faster than that of any other neurological disorder [6,7]. Disordered speech is among the most common motor consequences: dysarthria, hypophonia, and related vocal-tract dysfunctions affect the large majority of patients over the disease course [8,9]. Quantitative acoustic analysis detects these changes even in early and untreated disease [10]. Nonlinear dysphonia measures separate disordered from healthy voices [11], and repeated noninvasive speech tests have supported research on low-burden remote monitoring and progression tracking [12–14]. Automated speech analysis has revealed prodromal changes in at-risk cohorts [15]. Machine-learning approaches to voice-based PD detection have accordingly expanded rapidly, as documented by recent systematic reviews [16,17]. A digital biomarker, however, is clinically credible only when its evaluation matches its intended unit of use: the patient.

Public PD speech datasets often contain multiple recordings per participant. When recordings are shuffled independently, subject-specific anatomical and acquisition signatures can occur on both sides of a record-wise split, and the evaluation then estimates recognition of another recording from a previously observed person rather than generalization to an unseen person. This failure mode is a specific instance of data leakage, a recognized driver of over-optimistic results across applied machine learning [18,19]. This effect has been demonstrated directly in clinical prediction: record-wise cross-validation can inflate accuracy substantially relative to subject-wise evaluation of the very same pipeline [20], a finding that prompted an extended methodological discussion [21], and subject characteristics can drive apparent performance even when splits look innocuous [22]. Related biases arise whenever model selection and error estimation share data [23,24], and cross-validation on small samples carries wide error bars even under favorable conditions [25–27]. The UCI-489 corpus analyzed here contains three dependent replications from each of 80 subjects [28]; replication-aware analysis has been proposed for exactly this cohort [29].

Methodologically, three research strands converge in this study. First, engineered speech descriptors: dysphonia, cepstral, wavelet, and tunable Q-factor wavelet transform (TQWT) descriptor families are established carriers of Parkinsonian voice information [12,30,31]. Second, subspace geometry: collections of related observations can be represented as points on the Grassmann manifold $\mathrm{Gr}\left(r, d\right)$, whose quotient structure removes arbitrary basis choices [32–34]; this representation underpins subspace-based learning in computer vision and pattern recognition [35,36], and kernels on such manifolds require care because Gaussian functions of arbitrary manifold distances are not automatically positive semidefinite (PSD) [37]. Third, kernel fusion: reproducing-kernel Hilbert space (RKHS) theory [38–40] and multiple-kernel learning [41–43] combine heterogeneous similarities while preserving convex support vector optimization. Class imbalance, finally, is routinely addressed by synthetic oversampling [44–46], whose behavior in high-dimensional feature spaces warrants caution [47]. These strands have rarely been combined under an evaluation protocol that treats the patient as the statistical unit.

Deep architectures, including convolutional, recurrent, and transformer-based models, are increasingly applied to voice-based PD detection and scale well when large corpora are available [16]. Smartphone-scale collection efforts such as mPower point in that direction [48,49], and technology roadmaps for PD anticipate exactly this kind of remote, high-frequency assessment [50–52]. The cohorts relevant to replicated-phonation screening, however, typically contain tens to a few hundred subjects. In this regime, the present study deliberately combines structured geometric priors with a convex, auditable classical learner instead of high-capacity end-to-end models: each component (representation, kernel, solver, and resampling step) has an explicit mathematical description that can be verified directly against its implementation.

VoxNeuro aligns the mathematical unit of analysis with the clinical unit of interest: each subject is represented by the complete set of repeated measurements. A fixed-rank representation based on singular value decomposition (SVD) encodes the joint orientation of the repetitions, while a Euclidean summary preserves feature-space location and within-subject variability. A positive-semidefinite direct-sum kernel combines the two views, and evaluation is subject-disjoint throughout (Figure 1). For imbalanced data, synthetic subjects are generated along manifold geodesics rather than by interpolating individual recording rows. The working hypothesis is twofold: (i) subject-disjoint evaluation answers a different and more clinically meaningful question than record-wise splitting; and (ii) the geometric view carries predictive information beyond flexible Euclidean summaries, so that fusing the two views improves subject-level classification.

The principal contributions are as follows: (i) a formal subject-disjoint risk definition together with a quantitative account of the optimistic bias induced by record-wise splitting; (ii) a basis-invariant Grassmann representation of repeated speech; (iii) provably positive-semidefinite fusion of geometric and Euclidean kernels; (iv) geodesic minority oversampling coupled to the Euclidean view; and (v) deployment of a research-stage web application for guided speech capture and longitudinal result visualization. Evaluation across datasets uses leakage-safe, dataset-appropriate comparators, and on PD-252 a matched nonlinear Euclidean ablation with a radial basis function (RBF) kernel isolates the contribution of the fused geometric view.

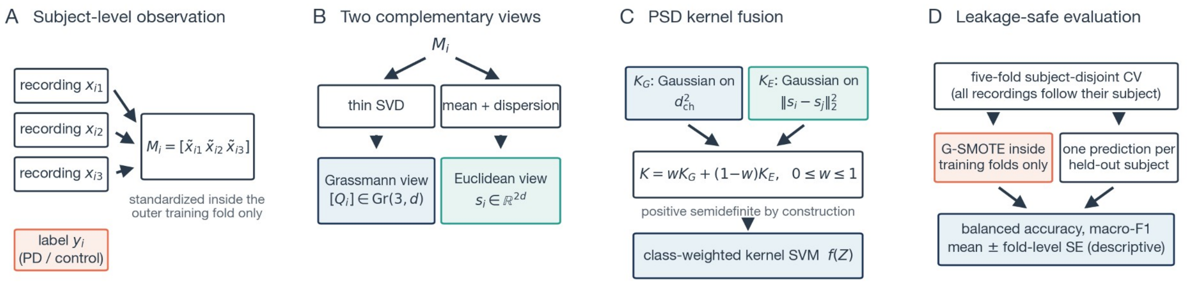

Figure 1. Schematic overview of the VoxNeuro framework. (**A**) Each subject contributes three repeated recordings and one diagnostic label; the recordings are standardized with statistics estimated inside the outer training fold only and stacked into a matrix M_i. (**B**) A thin singular value decomposition (SVD) yields the Grassmann view $[Q_i]$, while coordinate-wise mean and dispersion yield the Euclidean view s_i. (**C**) Gaussian kernels on the two views are combined into a positive-semidefinite (PSD) fused kernel feeding a class-weighted support vector machine (SVM). (**D**) Evaluation is five-fold subject-disjoint cross-validation (CV); Grassmann-geodesic synthetic minority oversampling (G-SMOTE) operates inside training folds only, and fold scores are summarized as mean ± standard error (SE). PD, Parkinson's disease.

2. Materials and Methods

2.1. Datasets

Two public, de-identified cohorts were analyzed (Table 1). The first, denoted UCI-489, comprises 80 subjects (40 PD, 40 control) with three recordings of sustained /a/ per subject, each recording described by 44 replicated acoustic descriptors and one dataset-provided binary sex/gender variable [28,29]. The second, denoted PD-252 [53], comprises 252 subjects (188 PD, 64 control) with three recordings per subject, described by 752 descriptors spanning baseline acoustic, vocal/time-frequency, Mel-frequency cepstral coefficient (MFCC) [54], wavelet, TQWT [55], and intrinsic mode function/empirical mode decomposition (IMF/EMD) [56] families [30,31]. It also includes one dataset-provided binary sex/gender variable, which constitutes the demographic family shown in Figure 8. The coded variable was retained as a prespecified model input and was never used to construct folds. This study was a secondary computational analysis of public data and involved no participant recruitment.

Table 1. Dataset regimes and evaluation design.

Attribute	UCI-489	PD-252
Subjects	80	252
PD / control	40 / 40	188 / 64
Recordings per subject	3	3
Modeled variables	45	753
Acoustic regime	Sustained /a/	Multi-family engineered features
Oversampling	Inactive	Minority control class only
Outer evaluation	Five-fold subject-disjoint cross-validation	

2.2. Subject-Disjoint Problem Formulation

Let $R_i = \{(x_{ij}, y_i) : j = 1, \ldots, m_i\}$ denote all recordings from subject i; Table 2 summarizes the notation used throughout. For outer fold k, let I_k^{tr} and I_k^{te} be the training and held-out subject-index sets. Leakage-safe evaluation requires

$$I_k^{\mathrm{tr}} \cap I_k^{\mathrm{te}} = \varnothing, I_k^{\mathrm{tr}} \cup I_k^{\mathrm{te}} = \{1, \ldots, n\}. \tag{1}$$

All recordings in R_i are assigned to the same partition as subject i (Figure 2A,B). The fold risk is evaluated once per held-out subject,

$$\hat{R}_k(f) = \frac{1}{|I_k^{\mathrm{te}}|} \sum_{i \in I_k^{\mathrm{te}}} \ell\left(f(Z_i), y_i\right), \tag{2}$$

where Z_i is the complete subject-level representation and ℓ is the classification loss. Equations (1) and (2) define the intended estimand: generalization to an unseen person rather than to an unseen row from a known person.

Table 2. Principal notation.

Symbol	Meaning
$x_{ij} \in \mathbb{R}^d$	feature vector of recording j of subject i
R_i, y_i	all recordings of subject i and its diagnostic label
$I_k^{\mathrm{tr}}, I_k^{\mathrm{te}}$	training and held-out subject-index sets of outer fold k
M_i	standardized $d \times 3$ recording matrix of subject i
Q_i, $[Q_i]$	orthonormal basis from the thin SVD and its subspace class
$\mathrm{Gr}(r,d)$, $\mathrm{St}(r,d)$	Grassmann and Stiefel manifolds of rank r in $\mathbb{R}^d$
$P_i = Q_i Q_i^\top$	orthogonal projector representing $[Q_i]$
$\theta_{ij,q}$, d_{ch}	principal angles and chordal distance between subspaces
$s_i \in \mathbb{R}^{2d}$	Euclidean mean–dispersion summary of subject i
K_G, K_E, K; w	Grassmann, Euclidean, and fused kernels; fusion weight
γ_G, γ_E	Gaussian bandwidths of the two views
$Q_{\mathrm{syn}}(t), s_{\mathrm{syn}}(t)$	geodesic and Euclidean interpolants of G-SMOTE
C_{y_i}	class-weighted SVM box constraint for class y_i
F, $\overline{m}$, SE	number of outer folds, fold-mean metric, standard error
I_{gk}, π_g	fold-level permutation importance and permutation of family g
h, A, $R_{\mathrm{seen}}, R_{\mathrm{new}}, R_{\mathrm{row}}$	record-level classifier, leakage event, and associated risks

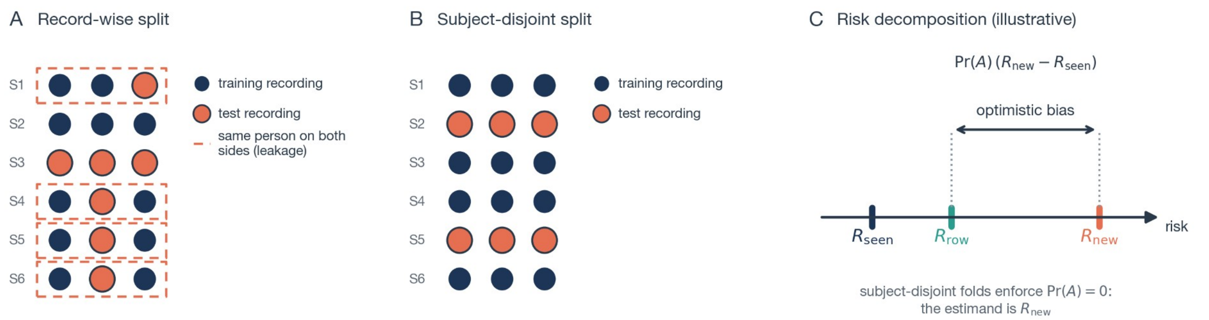

Figure 2. Schematic illustration of evaluation leakage with repeated recordings. (**A**) A record-wise split assigns individual recordings to training or test sets, so several subjects (dashed outlines) appear on both sides and the test set is not independent of training identities. (**B**) A subject-disjoint split assigns all recordings of a subject jointly. (**C**) Consequence of Equation (18): the record-wise estimate R_{row} interpolates between the easier seen-person risk R_{seen} and the deployment-relevant unseen-person risk R_{new}, drawn for an illustrative $\Pr(A) = 0.65$; the arrow marks the optimistic bias of magnitude $\Pr(A)(R_{\mathrm{new}} - R_{\mathrm{seen}})$.

2.3. Standardization and Subject Representation

For subject i, the three acoustic vectors $x_{ij} \in \mathbb{R}^d$ are standardized using the mean vector μ_k and scale vector estimated only from outer training fold k. Under the implemented standardization procedure, a zero-variance training coordinate is assigned unit scale:

$$\begin{aligned} \tilde{x}_{ij} &= D_k^{-1}(x_{ij} - \mu_k), D_k = \mathrm{diag}(\sigma_k^{\cdot}), \\ M_i &= [\tilde{x}_{i1} \tilde{x}_{i2} \tilde{x}_{i3}]. \end{aligned} \quad (3)$$

where $\sigma_k^{\cdot}$ denotes the training-fold scale vector after zero-variance handling. A thin SVD [57] gives

$$\begin{aligned} M_i &= U_i \Sigma_i V_i^{\top}, Q_i = U_i(:, 1:r), \\ Q_i^{\top} Q_i &= I_r, [Q_i] \in \mathrm{Gr}(r, d). \end{aligned} \quad (4)$$

Throughout, the rank is fixed at $r = 3$, matching the three repeated recordings per subject. The basis is not unique: Q_i and $Q_i O$, with O in the orthogonal group $\mathrm{O}(r)$, represent the same subspace. This quotient structure removes coordinate choices within the retained SVD basis (Figure 3A). In the official releases, one subject per cohort (CONT-36 in UCI-489; subject 37 in PD-252) contains byte-identical repeated recordings, so its matrix has numerical rank two and Equation (4) retains the orthonormal completion supplied by the SVD as the third basis vector; the reference implementation accepts such subjects only under an explicit opt-in flag.

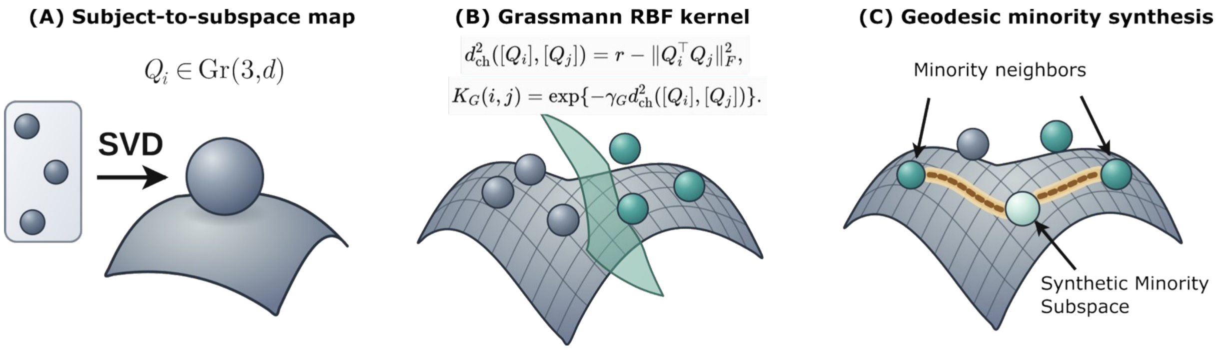

Figure 3. Geometric interpretation of the VoxNeuro representation, adapted from the project poster. (**A**) Three repeated recordings are mapped by a thin SVD to a fixed-rank representation on Gr(3,d). (**B**) Squared chordal distance and the associated Gaussian Grassmann radial-basis-function kernel compare subjects independently of the arbitrary orthonormal basis selected for each subspace. (**C**) Minority subjects are synthesized along a Grassmann geodesic rather than by row-wise Euclidean interpolation.

2.4. Principal Angles and Projection Geometry

Principal angles between two subject subspaces (Figure 4A) are defined by [32,58]

$$Q_i^{\top} Q_j = A_{ij} \cos(\Theta_{ij}) B_{ij}^{\top}, \quad \Theta_{ij} = \mathrm{diag}(\theta_{ij,1}, \ldots, \theta_{ij,r}). \quad (5)$$

where A_{ij} and B_{ij} are orthogonal. With the projector $P_i = Q_i Q_i^{\top}$, the squared chordal distance is given by [33,59]

$$\begin{aligned} d^2_{\mathrm{ch}}([Q_i],[Q_j]) &= ½\|P_i - P_j\|_F^2 \\ &= r - \|Q_i^{\top} Q_j\|_F^2 = \sum_{q=1}^{r} \sin^2 \theta_{ij,q}. \end{aligned} \quad (6)$$

Because $(Q_i O)(Q_i O)^{\top} = P_i$, the distance is basis invariant (Figure 4B). Vectorizing P_i maps the chordal distance to a Euclidean distance in projector space, up to a constant factor $\sqrt{2}$ absorbed by the bandwidth, thereby providing a Euclidean embedding for the Gaussian Grassmann kernel.

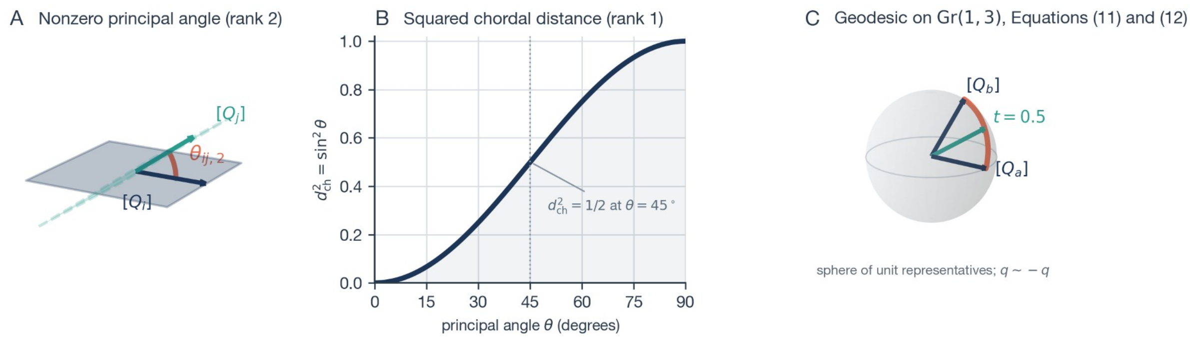

Figure 4. The geometry behind Equations (5), (6), (11), and (12), drawn for illustrative toy subspaces. (**A**) Two planes in $\mathbb{R}^3$ that share one direction, so the first principal angle is zero and the drawn angle is the second principal angle $\theta_{ij,2}$. (**B**) Exact rank-one relation between a principal angle and the squared chordal distance, $d^2_{\mathrm{ch}} = \sin^2\theta$. (**C**) A geodesic path computed with Equations (11) and (12) between two one-dimensional subspaces $[Q_a]$ and $[Q_b]$; every interpolated frame in the drawn path satisfies $Q_{\mathrm{syn}}(t)^\top Q_{\mathrm{syn}}(t) = I_r$ to machine precision, consistent with Proposition 2.

2.5. Complementary Euclidean View

The complementary Euclidean view retains the coordinate-wise mean and dispersion:

$$\begin{aligned} \mu_i &= ⅓\sum_{j=1}^{3}\tilde{x}_{ij}, \sigma_i = [⅓\sum_{j=1}^{3}(\tilde{x}_{ij}-\mu_i)^{\oslash 2}]^{1/2}, \\ s_i &= [\mu_i^\top \sigma_i^\top]^\top \in \mathbb{R}^{2d}. \end{aligned} \tag{7}$$

where $(\cdot)^{\oslash 2}$ and the square root are applied elementwise. Thus $[Q_i]$ encodes the orientation of the span of the standardized repetitions (M_i is not centered within subject, so this span generally includes the mean direction), whereas s_i retains coordinate-level location and dispersion.

2.6. Direct-Sum Kernel and Guarantee of Positive Semidefiniteness

Let ϕ_G and ϕ_E be feature maps for Gaussian kernels on the projection and Euclidean views [38,39]. With fusion weight w, the joint representation is the Hilbert direct sum

$$\begin{aligned} \Phi(Z_i) &= \left(\sqrt{w}\,\phi_G([Q_i]), \sqrt{1-w}\,\phi_E(s_i)\right), \\ &0 \le w \le 1. \end{aligned} \tag{8}$$

which induces, for bandwidths $\gamma_G, \gamma_E > 0$,

$$\begin{aligned} K_G(i,j) &= \exp\{-\gamma_G d^2_{\mathrm{ch}}([Q_i],[Q_j])\}, \\ K_E(i,j) &= \exp\{-\gamma_E \|s_i - s_j\|_2^2\}, \\ K(i,j) &= w K_G(i,j) + (1-w) K_E(i,j). \end{aligned} \tag{9}$$

Proposition 1 *(basis invariance and positive semidefiniteness).* The kernel in Equation (9) is invariant to $Q_i \mapsto Q_i O$ and is PSD. Basis invariance follows from the projector representation in Equation (6). Each Gaussian block is PSD on its Euclidean embedding, and a nonnegative weighted sum of PSD kernels is PSD [39,41]; hence, for every $c \in \mathbb{R}^n$,

$$c^\top K c = w \; c^\top K_G c + (1-w) c^\top K_E c \ge 0. \tag{10}$$

Equivalently, $K(i,j) = \langle \Phi(Z_i), \Phi(Z_j) \rangle$ in the direct-sum reproducing-kernel Hilbert space. □ The projector embedding is essential: Gaussian kernels built on other manifold distances, such as the geodesic arc length, are not PSD for all bandwidths [37], whereas the chordal construction used here inherits positive semidefiniteness from its explicit Euclidean embedding.

2.7. Grassmann-Geodesic Synthetic Minority Oversampling

We refer to this project-specific manifold adaptation as Grassmann-geodesic SMOTE (G-SMOTE); the abbreviation does not denote the separate Geometric SMOTE algorithm of Euclidean imbalanced learning [60]. Following the synthetic-neighbor principle of standard SMOTE [44] and its descendants [45,46,61], for minority subjects a and b,

$$\begin{aligned} Q_a^\top Q_b &= U\cos(\Theta)V^\top, \\ W_{ab} &= (Q_b V - Q_a U\cos\Theta)(\sin\Theta)^\dagger, \end{aligned} \tag{11}$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. A shortest geodesic and its coupled Euclidean interpolation are defined as follows (Figure 4C):

$$\begin{aligned} Q_{\mathrm{syn}}(t) &= [Q_a U\cos(t\Theta) + W_{ab}\sin(t\Theta)]V^\top, \\ s_{\mathrm{syn}}(t) &= s_a + t(s_b - s_a), \quad t \sim U(0,1). \end{aligned} \tag{12}$$

Within each imbalanced training fold, the minority class was the control class. For each synthetic subject, one minority subject was sampled, and a partner was selected from that subject's five nearest minority neighbors under squared chordal distance. Synthetic subjects were generated until the two class counts were equal. The same sampled value of t was used for the Grassmann and Euclidean interpolations. For numerical stabilization, the implementation applied a QR factorization to $Q_{\mathrm{syn}}(t)$. The resulting invertible right basis change preserves the represented Grassmann point. No synthetic subjects were generated for balanced UCI-489. Interpolating whole-subject representations along the manifold also avoids creating synthetic recording rows and constrains partner choice by subspace geometry, which mitigates, though does not eliminate, the known hazards of oversampling in high-dimensional feature spaces, where synthetic points can amplify noise directions [47].

Proposition 2 *(manifold validity).* In exact arithmetic, the nonzero-angle columns of $Q_a U$ and W_{ab} are mutually orthonormal, while zero-angle directions remain in the cosine term. Consequently,

$$Q_{\mathrm{syn}}(t)^\top Q_{\mathrm{syn}}(t) = V[\cos^2(t\Theta) + \sin^2(t\Theta)]V^\top = I_r. \tag{13}$$

Therefore, $Q_{\mathrm{syn}}(t) \in \mathrm{St}(r,d)$, the Stiefel manifold of orthonormal $d\times r$ frames; it represents a valid Grassmann point. Its endpoints are equivalent to $[Q_a]$ and $[Q_b]$. □

2.8. Kernel Support Vector Machine, Metrics, and Complexity

For support vector machine (SVM) training [62], labels are recoded as $y_i \in \{-1,+1\}$, $Y = \mathrm{diag}(y_1,\ldots,y_n)$, and balanced class weighting sets $C_{y_i} = Cn/(2n_{y_i})$, where n_{y_i} is the number of training subjects in class y_i; this is the balanced heuristic popularized by standard open-source implementations [63]. When the training classes are exactly balanced, as on UCI-489 and after G-SMOTE parity on PD-252, these weights reduce to $C_{y_i} = C$; the weighting is therefore active only for the unaugmented class-weighted comparators on imbalanced PD-252. The class-weighted dual and the decision function with bias b are

$$\begin{aligned} \max_{\alpha} \quad & 1^\top\alpha - \tfrac{1}{2}\alpha^\top YKY\alpha, \\ \text{subject to} \quad & 0 \le \alpha_i \le C_{y_i},\ y^\top\alpha = 0, \\ f(Z) \quad & = \sum_{i=1}^{n} \alpha_i y_i K(Z_i, Z) + b. \end{aligned} \tag{14}$$

Balanced accuracy (BA) [64] and macro-F1 [65] weight diagnostic classes symmetrically:

$$\begin{aligned} \mathrm{BA} &= \tfrac{1}{2}(\mathrm{TPR} + \mathrm{TNR}), \\ \text{macro-F1} &= \tfrac{1}{2}(F1_{\mathrm{PD}} + F1_{\mathrm{Ctrl}}). \end{aligned} \tag{15}$$

where TPR is sensitivity, TNR is specificity, and $F1_{\mathrm{PD}}$ and $F1_{\mathrm{Ctrl}}$ are the class-wise F1 scores; correlation-based alternatives such as the Matthews coefficient behave similarly for class-balanced summaries of two-class performance [66]. For $F=5$ outer folds, a metric m is summarized by

$$\begin{aligned} \overline{m} &= \frac{1}{F}\sum_{k=1}^{F} m_k, \\ \mathrm{SE}(\overline{m}) &= \left[\frac{1}{F(F-1)}\sum_{k=1}^{F}(m_k-\overline{m})^2\right]^{1/2}. \end{aligned} \tag{16}$$

The reported standard error (SE) is a descriptive measure of fold-to-fold variability and is not an inferential confidence interval.

For n subjects and rank $r=3$, subject SVDs require $O(ndr^2)$, the Grassmann Gram matrix requires $O(n^2dr^2)$, and the Euclidean block requires $O(n^2d)$. These bounds exclude the kernel-SVM optimization and the additional G-SMOTE neighbor search, which can dominate as n grows; all components remain tractable for the present cohort sizes $n=80$ and $n=252$.

2.9. Formal Effect of Record-Wise Leakage

The mechanism illustrated in Figure 2 can be stated formally, following record-wise versus subject-wise analyses in clinical machine learning [18,20,21]. Let (X,y) be a random validation recording and its label, let A be the event that the recording belongs to a person represented in training, and let h be a record-level classifier. With A^c the complement of A, define

$$\begin{aligned} R_{\mathrm{seen}} &= \mathrm{E}[\ell(h(X),y)\mid A], \\ R_{\mathrm{new}} &= \mathrm{E}[\ell(h(X),y)\mid A^c]. \end{aligned} \tag{17}$$

A record-wise split then estimates

$$\begin{aligned} R_{\mathrm{row}} &= \Pr(A)R_{\mathrm{seen}} + [1-\Pr(A)]R_{\mathrm{new}}, \\ R_{\mathrm{row}} - R_{\mathrm{new}} &= \Pr(A)(R_{\mathrm{seen}} - R_{\mathrm{new}}). \end{aligned} \tag{18}$$

If repeated recordings from previously observed people are easier than recordings from unseen people, then $R_{\mathrm{seen}} < R_{\mathrm{new}}$, and the record-wise estimate is optimistically biased in direct proportion to $\Pr(A)$ (Figure 2C). Subject-disjoint folds enforce $\Pr(A)=0$ by construction.

2.10. Experimental Protocol

Outer folds were generated by five-fold stratified subject-level cross-validation with shuffling and a random seed of 42 (Figure 5A). For each outer fold, the recording-level standardizer was fitted only to recordings from training subjects. Each Gaussian bandwidth was estimated from the positive squared pairwise distances δ^2 of the corresponding view over the training subjects, after G-SMOTE augmentation on PD-252, as $\gamma=[2\ \mathrm{median}(\delta^2)+10^{-12}]^{-1}$. Rank $r=3$, fusion weight $w=0.5$, and SVM penalty $C=1$ were fixed across datasets; no per-dataset hyperparameter search was performed, which removes the model-selection bias that shared tuning and testing data would introduce [23,24]. Because UCI-489 is balanced, no oversampling was applied there. On PD-252, the comparator set additionally included SMOTE plus logistic regression and a matched Euclidean RBF SVM using identical augmentation restricted to the training fold, so that the fused model and its nonlinear Euclidean ablation differed only in the geometric view (Figure 5B). All methods were implemented in Python, and the implementation is available in the project repository (see the Data Availability Statement). Table 3 summarizes all fixed protocol choices.

The comparator set was chosen to support an informative evaluation. Class-weighted logistic regression and class-weighted linear SVM probe whether simple decision boundaries on the Euclidean summary already suffice; SMOTE plus logistic regression tests a standard imbalanced-learning remedy [44]; and the matched Euclidean RBF SVM with identical G-SMOTE augmentation is the critical ablation, because it shares the summary features, the resampling, and the kernel family of the fused model while omitting only the Grassmann kernel term; the shared augmentation itself selects partners by chordal distance (Section 2.7). Any advantage of the fused model over this ablation is therefore attributable to the geometric kernel component, conditional on the shared augmentation, rather than to nonlinearity or resampling. On PD-252, G-SMOTE randomly samples anchor subjects and neighbors and draws interpolation coefficients, so performance varies with the augmentation seed. To isolate this source of variability, the fixed five-fold evaluation was repeated for 24 distinct augmentation seeds while the outer-fold assignments, preprocessing, hyperparameters, and all non-augmentation settings were held fixed; within each seed and fold, the fused model and the matched Euclidean RBF ablation used the same augmented training set, so their difference is paired at the seed level. The random state of the original PD-252 augmentation realization was not retained, so that specific run cannot be regenerated exactly from a seed; the seed analysis of Section 3.3 therefore characterizes where it lies within the distribution of realizations.

Table 3. Frozen experimental protocol. All learned components are fitted inside the outer training folds only.

Component	**Setting**
Outer evaluation	five-fold stratified subject-level cross-validation, shuffled, seed 42
Standardization	per-coordinate mean / scale from training recordings; zero-variance coordinates assigned unit scale
Subject representation	rank $r = 3$ thin SVD (Grassmann view) + mean / dispersion vector (Euclidean view)
Kernel bandwidths	$\gamma = [2\ \ \mathrm{median}(\delta^2) + 10^{-12}]^{-1}$ from positive squared pairwise training distances (augmented training set on PD-252)
Fusion weight / penalty	$w = 0.5$; class-weighted SVM with $C = 1$; no per-dataset tuning
Oversampling	none on UCI-489; G-SMOTE to class parity on PD-252, five nearest minority neighbors under chordal distance
Comparators	class-weighted logistic regression; class-weighted linear SVM; SMOTE + logistic regression and matched Euclidean RBF SVM on PD-252
Metrics	balanced accuracy and macro-F1, mean ± fold-level SE; pooled out-of-fold confusion counts
Sensitivity analysis	held-out permutation of Euclidean summary blocks, no refitting

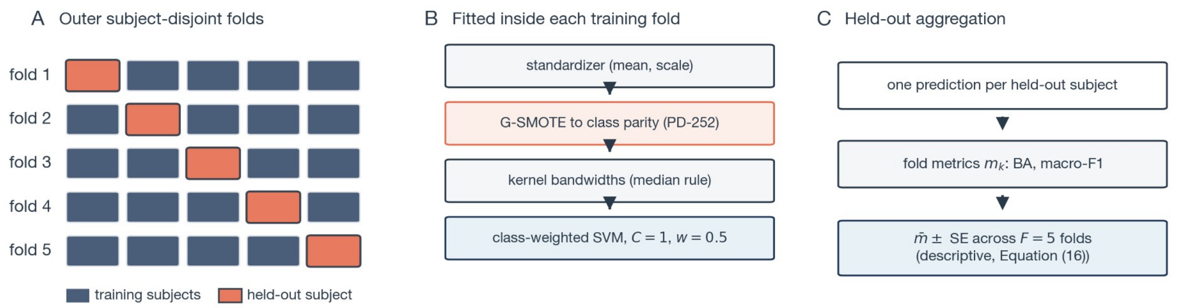

Figure 5. Evaluation protocol. (**A**) Five-fold subject-disjoint partition; in each outer fold one subject block (orange) is held out and never influences any fitted component. (**B**) Everything that learns from data—the standardizer, G-SMOTE, the kernel bandwidths, and the SVM—is fitted inside the training fold only. (**C**) Exactly one prediction is issued per held-out subject, fold metrics are computed at the subject level, and Equation (16) aggregates them descriptively.

2.11. Feature-Family Permutation Sensitivity

For feature family g, the fold-level decrease in performance and its summary are

$$
\begin{aligned}
I_{gk} &= m_k - m_k^{\pi_g}, \hat{I}_g = \frac{1}{F}\sum_{k=1}^{F} I_{gk}, \\
\mathrm{SE}(\hat{I}_g) &= \left[\frac{1}{F(F-1)}\sum_{k=1}^{F}(I_{gk}-\hat{I}_g)^2\right]^{1/2}.
\end{aligned}
\tag{19}
$$

Sensitivity of the Euclidean summary was evaluated on each held-out fold without refitting. The coordinates of s_i associated with family g were jointly permuted across held-out subjects while Q_i and all other summary coordinates were held fixed. In Equation (19), m_k is the original held-out macro-F1 and $m_k^{\pi_g}$ is macro-F1 after permutation. Each family and fold used a single permutation realization, so fold-level variability also includes permutation-draw noise.

3. Results

3.1. Performance Across Datasets

Figure 6 shows that the fused model attained the highest mean balanced accuracy and macro-F1 among the audited comparators in both datasets; Table 4 summarizes the principal scores. On UCI-489, Grassmann–Euclidean fusion achieved a balanced accuracy of 0.850 with an SE of 0.042 and a macro-F1 of 0.847 with an SE of 0.043. Its mean scores were 0.075 balanced-accuracy points and 0.077 macro-F1 points higher than those of class-weighted logistic regression, the strongest audited UCI-489 baseline. On PD-252, fusion with G-SMOTE achieved a balanced accuracy of 0.749 with an SE of 0.039 and a macro-F1 of 0.760 with an SE of 0.035. The matched Euclidean RBF model with identical augmentation achieved a balanced accuracy of 0.736 with an SE of 0.035 and a macro-F1 of 0.735 with an SE of 0.027. Thus, the fused model's mean scores were 0.013 balanced-accuracy points and 0.025 macro-F1 points higher than those of the matched nonlinear Euclidean ablation. These are descriptive cross-validation differences from a single G-SMOTE realization on PD-252 (Section 3.3); no inferential test of model superiority was performed.

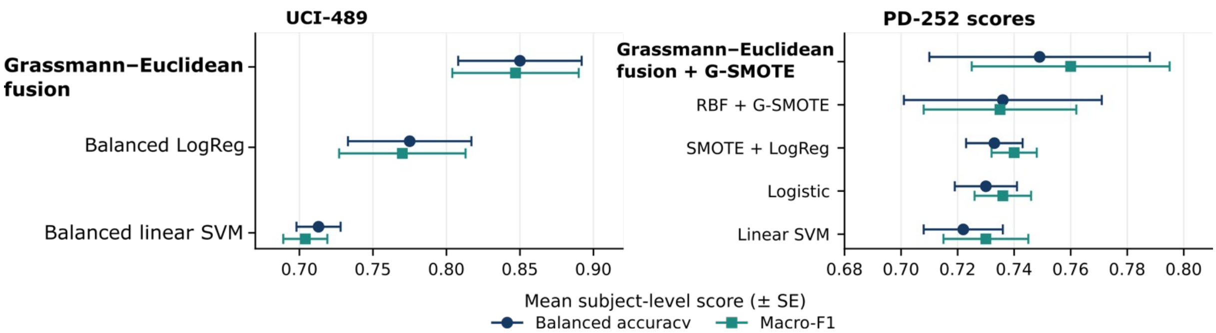

Figure 6. Subject-level speech results (left: balanced UCI-489; right: high-dimensional, imbalanced PD-252). Whiskers denote standard error across five outer folds. RBF + G-SMOTE is the matched nonlinear Euclidean ablation and SMOTE + LogReg is SMOTE with logistic regression; Balanced LogReg and Logistic denote class-weighted logistic regression, and Balanced linear SVM and Linear SVM denote class-weighted linear SVM. PD-252 values for the fused model and the matched Euclidean ablation are from the original single G-SMOTE augmentation realization; paired 24-seed summaries are reported in Table 6.

Table 4. Headline subject-level performance (mean ± fold-level standard error over five subject-disjoint folds). The strongest audited baseline is shown for each dataset; all comparator summary estimates appear in Figure 6. PD-252 rows report the original single G-SMOTE realization; seed sensitivity is quantified in Table 6.

Dataset	Model	Balanced acc.	Macro-F1
UCI-489	Grassmann–Euclidean fusion	0.850 ± 0.042	0.847 ± 0.043
UCI-489	Class-weighted logistic regression	0.775 ± 0.042	0.770 ± 0.043
PD-252	Fusion + G-SMOTE	0.749 ± 0.039	0.760 ± 0.035
PD-252	Euclidean RBF + G-SMOTE (matched ablation)	0.736 ± 0.035	0.735 ± 0.027

3.2. Operating Characteristics

The UCI-489 fused model produced 34 true-negative (control), six false-positive, six false-negative, and 34 true-positive (PD) classifications, yielding sensitivity and specificity of 0.850. On PD-252, under the original single G-SMOTE realization, the pooled confusion matrix contained 173 true-positive (PD), 15 false-negative, 37 true-negative (control), and 27 false-positive classifications, corresponding to sensitivity 0.920, specificity 0.578, positive predictive value 0.865, and negative predictive value 0.712 (Figure 7, Table 5). Sensitivity and specificity are prevalence-free, whereas PPV and NPV depend on this cohort's 188/64 composition and the default decision threshold and do not transport to lower-prevalence screening settings. Rates computed from pooled confusion matrices are descriptive and need not equal the arithmetic mean of fold-level metrics. The PD-252 profile is sensitive but would create a meaningful referral burden; this profile supports evaluation as a research-stage screening aid rather than autonomous diagnosis.

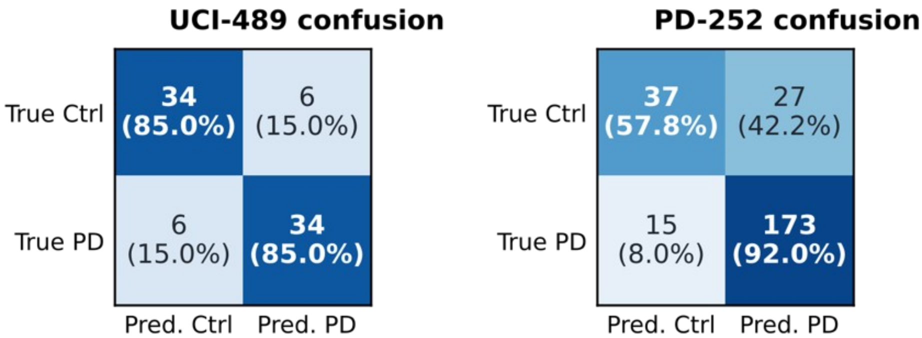

Figure 7. Pooled out-of-fold confusion matrices for UCI-489 (left) and PD-252 (right), containing one prediction per subject; percentages are normalized within the true class. PD-252 counts are from the original single G-SMOTE augmentation realization.

Table 5. Pooled out-of-fold operating characteristics computed from the confusion counts of Figure 7 (UCI-489: TP = 34, FN = 6, TN = 34, FP = 6; PD-252: TP = 173, FN = 15, TN = 37, FP = 27). TP, true positive; FN, false negative; TN, true negative; FP, false positive; PPV, positive predictive value; NPV, negative predictive value.

Dataset	Sensitivity	Specificity	PPV	NPV
UCI-489	0.850	0.850	0.850	0.850
PD-252	0.920	0.578	0.865	0.712

3.3. Sensitivity to the G-SMOTE Augmentation Seed

The PD-252 scores in Table 4 were obtained from one G-SMOTE augmentation realization. The paired analysis of Section 2.10 characterizes the sensitivity of both models and of their difference to the augmentation seed (Table 6). Across 24 seed runs, balanced accuracy was 0.739 ± 0.012 for the fused model and 0.731 ± 0.013 for the matched Euclidean ablation, and macro-F1 was 0.752 ± 0.013 and 0.729 ± 0.013, respectively (mean ± standard deviation across seeds). The paired macro-F1 difference averaged +0.023 ± 0.015 and was positive in 23 of 24 runs; the original realization's difference of +0.025 lies within this distribution. The paired balanced-accuracy difference averaged +0.009 ± 0.014, was positive in 16 of 24 runs, and reversed sign in 8 of 24. The original model scores lie within their corresponding empirical seed ranges, 0.6–0.8 seed standard deviations above the seed means, and the reference implementation's default seed yields 0.731 and 0.749 for the fused model, likewise within these ranges; this run and its out-of-fold predictions are archived in the project repository. Under seed means, the fused model again attained the highest mean balanced accuracy and macro-F1 among the audited PD-252 comparators; the two class-weighted comparators without augmentation are unaffected by this seed, and SMOTE plus logistic regression is represented by its single archived realization. Thus, conditional on the fixed outer folds, the macro-F1 margin was directionally consistent in all but one seed run, whereas the balanced-accuracy margin can be reversed by individual draws.

Table 6. Sensitivity of the PD-252 comparison to the G-SMOTE augmentation seed: paired evaluation of the fused model and the matched Euclidean RBF ablation across 24 distinct seeds with outer folds, preprocessing, and hyperparameters held fixed. Values are means ± standard deviations (SD) across seeds; Δ denotes the paired difference (fused minus ablation) computed within each seed, summarized by its mean ± SD, median [interquartile range, IQR], and sign count.

Metric	Fused	Ablation	Δ mean ± SD	Δ median [IQR]	Δ > 0
Balanced accuracy	0.739 ± 0.012	0.731 ± 0.013	+0.009 ± 0.014	+0.008 [−0.002, +0.018]	16/24
Macro-F1	0.752 ± 0.013	0.729 ± 0.013	+0.023 ± 0.015	+0.020 [+0.012, +0.034]	23/24

Each seed-level value is the five-fold cross-validation point estimate under fixed outer folds; standard deviations describe variation across the 24 augmentation seeds and are distinct from the fold-level standard errors of Table 4. Δ ranges: balanced accuracy [−0.013, +0.042]; macro-F1 [−0.001, +0.061]. Per-seed values are archived in the project repository.

3.4. Feature-Family Sensitivity

The TQWT summary block produced the largest mean decrease in held-out macro-F1, at 0.143 with an SE of 0.037, followed by the MFCC summary block, at 0.050 with an SE of 0.017 (Figure 8). The vocal/time-frequency, wavelet, IMF/EMD, and demographic families produced mean decreases near zero. These exploratory values measure dependence of the fitted model on the Euclidean summary coordinates under correlated features; they do not establish biological causality or total dependence of both model views. The prominence of the TQWT family is consistent with the descriptor engineering of the source dataset, where tunable Q-factor features were introduced precisely for Parkinsonian voice [31,55].

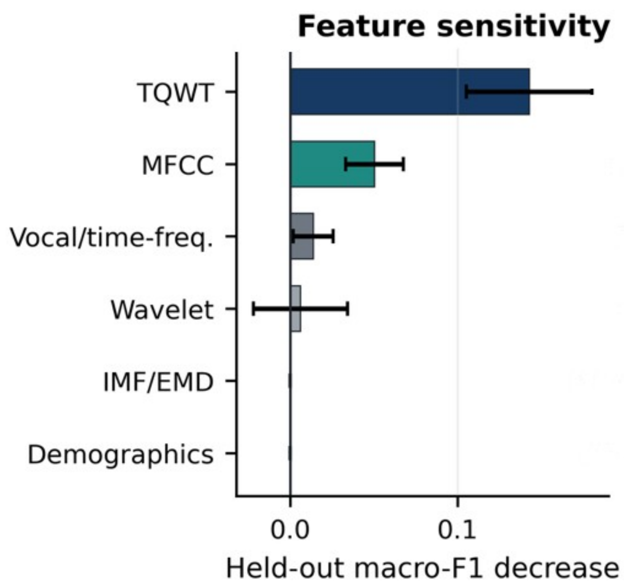

Figure 8. PD-252 feature-family permutation sensitivity on the Euclidean summary view. Positive values denote a decrease in held-out macro-F1 after permutation, and whiskers denote fold-level standard error. TQWT denotes tunable Q-factor wavelet transform; MFCC denotes Mel-frequency cepstral coefficients; IMF/EMD denotes intrinsic mode function/empirical mode decomposition; Demographics denotes the dataset-provided sex/gender covariate.

4. Discussion

The central methodological contribution is the alignment of the evaluation unit with the clinical unit. Subject-disjoint validation asks whether a model generalizes to an unseen person. The Grassmann representation uses the dependence among repeated recordings rather than treating the recordings as independent replicates. Basis invariance removes arbitrary coordinate choices, while the Euclidean view restores the coordinate-level location and dispersion magnitudes that a pure subspace representation discards.

Across both datasets, the fused model attained the highest mean balanced accuracy and macro-F1 among the audited comparators, and on PD-252 it exceeded the matched Euclidean RBF ablation with G-SMOTE. This establishes the highest mean performance within the audited comparator set; it does not establish universal or statistically significant superiority over all learning algorithms. The seed-sensitivity analysis of Section 3.3 sharpens the ablation comparison: the paired macro-F1 difference was positive in 23 of 24 augmentation runs, whereas the balanced-accuracy difference was positive in 16 of 24 and reversed in 8. These counts characterize sensitivity to the augmentation randomization under fixed outer folds; they are not population-level inference about model superiority. Equations (17) and (18) make the mechanism behind the design explicit. Whenever repeated recordings from previously observed people are easier to classify than recordings from unseen people, record-wise scores are optimistically biased in direct proportion to the probability that a validation recording belongs to an already-observed person. Subject-disjoint folds remove that term entirely. Reported subject-level scores are therefore expected to be lower than record-wise figures reported on the same corpora, but they answer the question a screening deployment actually poses [20,22].

An evaluation-focused perspective also clarifies how these results should be interpreted in relation to the wider literature. Reported accuracies for voice-based PD detection vary widely across studies, and validation strategy is a recognized source of that variation [16,17]; record-wise protocols remain common on corpora that contain repeated recordings per participant, including the two analyzed here [28,29,31]. Balanced accuracy and macro-F1 were used because both cohorts were evaluated at the subject level and PD-252 had unequal class sizes. Both metrics weight the diagnostic classes symmetrically rather than rewarding majority-class predictions [64,65]. Fold-level standard errors are reported descriptively, and no significance test is claimed: five outer folds provide a descriptive summary of variability, not a basis for inferential comparison [25].

The deployed VoxNeuro web application (Figure 9) requires only a commodity microphone and a web browser, which keeps the barrier to repeated assessment low. This is precisely the kind of low-friction, high-frequency measurement that technology assessments of PD care have called for [50,51]. It is a research-stage prototype, not an autonomous diagnostic system or medical device, and no application-collected data were used in the analyses reported here. Prospective use requires explicit privacy, consent, security, accessibility conformance, calibration, and clinical governance procedures.

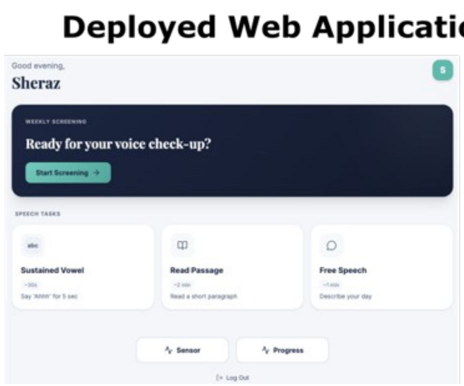

Figure 9. Deployed VoxNeuro web application for guided speech capture and longitudinal visualization of research-stage screening results. The application requires only a commodity microphone and a web browser; no application-collected data were used in the analyses reported here.

This study has limitations. The speech cohorts contain retrospective engineered features rather than prospectively acquired raw audio. The study included no probability calibration [67,68], decision-curve analysis [69], independent clinical test cohort, device-shift evaluation, language-shift audit, or prespecified subgroup analysis. Because high PD-252 sensitivity coexists with modest specificity, a credible system should support referral and longitudinal review rather than replace neurological assessment. Future work should preregister preprocessing and thresholds and report in accordance with structured guidance for clinical prediction models [70,71]; collect repeated raw speech prospectively; and evaluate calibration, test–retest reliability, fairness, and clinician-facing decision consequences. The two-view construction extends naturally to additional recording tasks and views through further positive-semidefinite kernel terms, with weights selected by nested multiple-kernel optimization [42,43].

Reproducibility safeguards included fixed subject-disjoint folds, training-fold-only preprocessing, fixed hyperparameters, saved fold-level metrics, pooled confusion counts, and restriction of reported results to audited comparator implementations. The project repository archives subject-level out-of-fold predictions with fold identifiers for UCI-489, a pinned dependency environment, and the seed-sensitivity script with its 24 per-seed paired metrics, so that the UCI-489 aggregates of Tables 4 and 5 can be regenerated exactly and the seed-level summaries of Table 6 can be reconstructed; the original PD-252 realization is situated within the archived seed distribution (Section 3.3).

5. Conclusions

VoxNeuro provides a leakage-safe framework for repeated Parkinsonian speech. Each subject is represented by a Grassmann subspace and a Euclidean mean–variability vector. The corresponding positive-semidefinite kernels are fused, and minority subjects are synthesized along manifold geodesics only within training folds. The fused model attained the highest mean balanced accuracy and macro-F1 among the audited comparators on UCI-489 (0.850 and 0.847) and, across 24 paired augmentation seeds, on PD-252 (0.739 ± 0.012 and 0.752 ± 0.013), where its macro-F1 exceeded that of the matched Euclidean RBF ablation in 23 of 24 draws. The deployed web application implements guided capture and longitudinal visualization as a research-stage prototype. Clinically credible digital biomarkers require expressive mathematics, auditable implementations, transparent uncertainty, and evaluation designs that generalize to unseen people.

Author Contributions: Conceptualization, L.K.; methodology, L.K.; software, L.K.; validation, L.K. and M.W.; formal analysis, L.K.; investigation, L.K. and M.W.; data curation, L.K.; writing—original draft preparation, L.K.; writing—review and editing, L.K. and M.W.; visualization, L.K.; project administration, L.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were not required for this study, as it consisted exclusively of a secondary computational analysis of publicly available, fully de-identified datasets; no participants were recruited, and no new human data were collected.

Informed Consent Statement: Not applicable. The study analyzed publicly available, de-identified datasets and did not involve direct human participation; informed consent and ethics procedures for the original data collections are documented in the source publications cited in Section 2.1.

Data Availability Statement: The datasets analyzed in this study are openly available from the UCI Machine Learning Repository: the Parkinson Dataset with Replicated Acoustic Features (UCI-489; UCI dataset 489, doi:10.24432/C5701F) and the Parkinson's Disease Classification dataset (PD-252; UCI dataset 470, doi:10.24432/C5MS4X). Source code is available at https://github.com/khan-laiba/voxneuro; the repository also archives subject-level out-of-fold prediction files for UCI-489, the pinned dependency environment, and the per-seed outputs of the PD-252 seed-sensitivity analysis. The deployed research-stage web application is available at https://voxneuro.org. No new primary data were collected in this study.

Acknowledgments: During the preparation of this manuscript, the authors used Claude (Anthropic; Claude Opus) to draft and edit manuscript text, format the manuscript, and produce the schematic illustrations in Figures 1, 2, 4, and 5, and GPT-5.6 Sol (OpenAI; via the Codex CLI) to review manuscript language and figure design. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Bloem, B.R.; Okun, M.S.; Klein, C. Parkinson's disease. *Lancet* **2021**, *397*, 2284–2303.
2. Kalia, L.V.; Lang, A.E. Parkinson's disease. *Lancet* **2015**, *386*, 896–912.
3. Poewe, W.; Seppi, K.; Tanner, C.M.; Halliday, G.M.; Brundin, P.; Volkmann, J.; Schrag, A.-E.; Lang, A.E. Parkinson disease. *Nat. Rev. Dis. Primers* **2017**, *3*, 17013.
4. Postuma, R.B.; Berg, D.; Stern, M.; Poewe, W.; Olanow, C.W.; Oertel, W.; Obeso, J.; Marek, K.; Litvan, I.; Lang, A.E.; et al. MDS clinical diagnostic criteria for Parkinson's disease. *Mov. Disord.* **2015**, *30*, 1591–1601.
5. Armstrong, M.J.; Okun, M.S. Diagnosis and treatment of Parkinson disease: A review. *JAMA* **2020**, *323*, 548–560.
6. GBD 2016 Parkinson's Disease Collaborators. Global, regional, and national burden of Parkinson's disease, 1990 –2016: A systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol.* **2018**, *17*, 939–953.
7. Dorsey, E.R.; Bloem, B.R. The Parkinson pandemic—A call to action. *JAMA Neurol.* **2018**, *75*, 9–10.
8. Logemann, J.A.; Fisher, H.B.; Boshes, B.; Blonsky, E.R. Frequency and cooccurrence of vocal tract dysfunctions in the speech of a large sample of Parkinson patients. *J. Speech Hear. Disord.* **1978**, *43*, 47–57.
9. Ho, A.K.; Iansek, R.; Marigliani, C.; Bradshaw, J.L.; Gates, S. Speech impairment in a large sample of patients with Parkinson's disease. *Behav. Neurol.* **1998**, *11*, 131–137.
10. Rusz, J.; Cmejla, R.; Ruzickova, H.; Ruzicka, E. Quantitative acoustic measurements for characterization of speech and voice disorders in early untreated Parkinson's disease. *J. Acoust. Soc. Am.* **2011**, *129*, 350–367.
11. Little, M.A.; McSharry, P.E.; Roberts, S.J.; Costello, D.A.E.; Moroz, I.M. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *Biomed. Eng. Online* **2007**, *6*, 23.
12. Little, M.A.; McSharry, P.E.; Hunter, E.J.; Spielman, J.; Ramig, L.O. Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. *IEEE Trans. Biomed. Eng.* **2009**, *56*, 1015–1022.
13. Tsanas, A.; Little, M.A.; McSharry, P.E.; Ramig, L.O. Accurate telemonitoring of Parkinson's disease progression by noninvasive speech tests. *IEEE Trans. Biomed. Eng.* **2010**, *57*, 884–893.
14. Tsanas, A.; Little, M.A.; McSharry, P.E.; Spielman, J.; Ramig, L.O. Novel speech signal processing algorithms for high-accuracy classification of Parkinson's disease. *IEEE Trans. Biomed. Eng.* **2012**, *59*, 1264–1271.
15. Hlavnička, J.; Čmejla, R.; Tykalová, T.; Šonka, K.; Růžička, E.; Rusz, J. Automated analysis of connected speech reveals early biomarkers of Parkinson's disease in patients with rapid eye movement sleep behaviour disorder. *Sci. Rep.* **2017**, *7*, 12.
16. Sedigh Malekroodi, H.; Lee, B.-i.; Yi, M. Voice-based detection of Parkinson's disease using machine and deep learning approaches: A systematic review. *Bioengineering* **2025**, *12*, 1279.
17. Moro-Velazquez, L.; Gomez-Garcia, J.A.; Arias-Londoño, J.D.; Dehak, N.; Godino-Llorente, J.I. Advances in Parkinson's disease detection and assessment using voice and speech: A review of the state-of-the-art. *Biomed. Signal Process. Control* **2021**, *66*, 102418.
18. Kaufman, S.; Rosset, S.; Perlich, C.; Stitelman, O. Leakage in data mining: Formulation, detection, and avoidance. *ACM Trans. Knowl. Discov. Data* **2012**, *6*, 15.
19. Kapoor, S.; Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **2023**, *4*, 100804.
20. Saeb, S.; Lonini, L.; Jayaraman, A.; Mohr, D.C.; Kording, K.P. The need to approximate the use-case in clinical machine learning. *Gigascience* **2017**, *6*, gix019.
21. Little, M.A.; Varoquaux, G.; Saeb, S.; Lonini, L.; Jayaraman, A.; Mohr, D.C.; Kording, K.P. Using and understanding cross-validation strategies. Perspectives on Saeb et al. *Gigascience* **2017**, *6*, gix020.
22. Chaibub Neto, E.; Pratap, A.; Perumal, T.M.; Tummalacherla, M.; Snyder, P.; Bot, B.M.; Trister, A.D.; Friend, S.H.; Mangravite, L.; Omberg, L. Detecting the impact of subject characteristics on machine learning-based diagnostic applications. *NPJ Digit. Med.* **2019**, *2*, 99.
23. Cawley, G.C.; Talbot, N.L.C. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J. Mach. Learn. Res.* **2010**, *11*, 2079–2107.
24. Varma, S.; Simon, R. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics* **2006**, *7*, 91.
25. Varoquaux, G. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage* **2018**, *180*, 68–77.
26. Arlot, S.; Celisse, A. A survey of cross-validation procedures for model selection. *Stat. Surv.* **2010**, *4*, 40–79.
27. Hastie, T.; Tibshirani, R.; Friedman, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed.; Springer: New York, NY, USA, 2009.
28. Pérez, C.J. *Parkinson Dataset with Replicated Acoustic Features*; UCI Machine Learning Repository, 2016. doi:10.24432/C5701F.

29. Naranjo, L.; Pérez, C.J.; Martín, J.; Campos-Roca, Y. A two-stage variable selection and classification approach for Parkinson's disease detection by using voice recording replications. *Comput. Methods Programs Biomed.* **2017**, *142*, 147–156.
30. Sakar, B.E.; Isenkul, M.E.; Sakar, C.O.; Sertbas, A.; Gurgen, F.; Delil, S.; Apaydin, H.; Kursun, O. Collection and analysis of a Parkinson speech dataset with multiple types of sound recordings. *IEEE J. Biomed. Health Inform.* **2013**, *17*, 828–834.
31. Sakar, C.O.; Serbes, G.; Gunduz, A.; Tunc, H.C.; Nizam, H.; Sakar, B.E.; Tutuncu, M.; Aydin, T.; Isenkul, M.E.; Apaydin, H. A comparative analysis of speech signal processing algorithms for Parkinson's disease classification and the use of the tunable Q-factor wavelet transform. *Appl. Soft Comput.* **2019**, *74*, 255–263.
32. Wong, Y.-C. Differential geometry of Grassmann manifolds. *Proc. Natl. Acad. Sci. USA* **1967**, *57*, 589–594.
33. Edelman, A.; Arias, T.A.; Smith, S.T. The geometry of algorithms with orthogonality constraints. *SIAM J. Matrix Anal. Appl.* **1998**, *20*, 303–353.
34. Absil, P.-A.; Mahony, R.; Sepulchre, R. *Optimization Algorithms on Matrix Manifolds*; Princeton University Press: Princeton, NJ, USA, 2008.
35. Hamm, J.; Lee, D.D. Grassmann discriminant analysis: A unifying view on subspace-based learning. In Proceedings of the 25th International Conference on Machine Learning, Helsinki, Finland, 5–9 July 2008; pp. 376–383.
36. Turaga, P.; Veeraraghavan, A.; Srivastava, A.; Chellappa, R. Statistical computations on Grassmann and Stiefel manifolds for image and video-based recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* **2011**, *33*, 2273–2286.
37. Jayasumana, S.; Hartley, R.; Salzmann, M.; Li, H.; Harandi, M. Kernel methods on Riemannian manifolds with Gaussian RBF kernels. *IEEE Trans. Pattern Anal. Mach. Intell.* **2015**, *37*, 2464–2477.
38. Aronszajn, N. Theory of reproducing kernels. *Trans. Am. Math. Soc.* **1950**, *68*, 337–404.
39. Schölkopf, B.; Smola, A.J. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*; MIT Press: Cambridge, MA, USA, 2002.
40. Shawe-Taylor, J.; Cristianini, N. *Kernel Methods for Pattern Analysis*; Cambridge University Press: Cambridge, UK, 2004.
41. Lanckriet, G.R.G.; Cristianini, N.; Bartlett, P.; El Ghaoui, L.; Jordan, M.I. Learning the kernel matrix with semidefinite programming. *J. Mach. Learn. Res.* **2004**, *5*, 27–72.
42. Gönen, M.; Alpaydın, E. Multiple kernel learning algorithms. *J. Mach. Learn. Res.* **2011**, *12*, 2211–2268.
43. Rakotomamonjy, A.; Bach, F.R.; Canu, S.; Grandvalet, Y. SimpleMKL. *J. Mach. Learn. Res.* **2008**, *9*, 2491–2521.
44. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357.
45. He, H.; Garcia, E.A. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* **2009**, *21*, 1263–1284.
46. Fernández, A.; García, S.; Herrera, F.; Chawla, N.V. SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *J. Artif. Intell. Res.* **2018**, *61*, 863–905.
47. Blagus, R.; Lusa, L. SMOTE for high-dimensional class-imbalanced data. *BMC Bioinformatics* **2013**, *14*, 106.
48. Bot, B.M.; Suver, C.; Neto, E.C.; Kellen, M.; Klein, A.; Bare, C.; Doerr, M.; Pratap, A.; Wilbanks, J.; Dorsey, E.R.; et al. The mPower study, Parkinson disease mobile data collected using ResearchKit. *Sci. Data* **2016**, *3*, 160011.
49. Arora, S.; Venkataraman, V.; Zhan, A.; Donohue, S.; Biglan, K.M.; Dorsey, E.R.; Little, M.A. Detecting and monitoring the symptoms of Parkinson's disease using smartphones: A pilot study. *Parkinsonism Relat. Disord.* **2015**, *21*, 650–653.
50. Espay, A.J.; Bonato, P.; Nahab, F.B.; Maetzler, W.; Dean, J.M.; Klucken, J.; Eskofier, B.M.; Merola, A.; Horak, F.; Lang, A.E.; et al. Technology in Parkinson's disease: Challenges and opportunities. *Mov. Disord.* **2016**, *31*, 1272–1282.
51. Dorsey, E.R.; Glidden, A.M.; Holloway, M.R.; Birbeck, G.L.; Schwamm, L.H. Teleneurology and mobile technologies: The future of neurological care. *Nat. Rev. Neurol.* **2018**, *14*, 285–297.
52. Topol, E.J. High-performance medicine: The convergence of human and artificial intelligence. *Nat. Med.* **2019**, *25*, 44–56.
53. Sakar, C.O.; Serbes, G.; Gunduz, A.; Nizam, H.; Sakar, B.E. *Parkinson's Disease Classification*; UCI Machine Learning Repository, 2018. doi:10.24432/C5MS4X.
54. Davis, S.B.; Mermelstein, P. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Trans. Acoust. Speech Signal Process.* **1980**, *28*, 357–366.
55. Selesnick, I.W. Wavelet transform with tunable Q-factor. *IEEE Trans. Signal Process.* **2011**, *59*, 3560–3575.
56. Huang, N.E.; Shen, Z.; Long, S.R.; Wu, M.C.; Shih, H.H.; Zheng, Q.; Yen, N.-C.; Tung, C.C.; Liu, H.H. The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. *Proc. R. Soc. Lond. A* **1998**, *454*, 903–995.

57. Golub, G.H.; Van Loan, C.F. *Matrix Computations*, 4th ed.; Johns Hopkins University Press: Baltimore, MD, USA, 2013.
58. Björck, Å.; Golub, G.H. Numerical methods for computing angles between linear subspaces. *Math. Comput.* **1973**, *27*, 579–594.
59. Chikuse, Y. *Statistics on Special Manifolds*; Springer: New York, NY, USA, 2003.
60. Douzas, G.; Bação, F. Geometric SMOTE: A geometrically enhanced drop-in replacement for SMOTE. *Inf. Sci.* **2019**, *501*, 118–135.
61. Batista, G.E.A.P.A.; Prati, R.C.; Monard, M.C. A study of the behavior of several methods for balancing machine learning training data. *ACM SIGKDD Explor. Newsl.* **2004**, *6*, 20–29.
62. Cortes, C.; Vapnik, V. Support-vector networks. *Mach. Learn.* **1995**, *20*, 273–297.
63. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
64. Brodersen, K.H.; Ong, C.S.; Stephan, K.E.; Buhmann, J.M. The balanced accuracy and its posterior distribution. In Proceedings of the 20th International Conference on Pattern Recognition, Istanbul, Turkey, 23–26 August 2010; pp. 3121–3124.
65. Sokolova, M.; Lapalme, G. A systematic analysis of performance measures for classification tasks. *Inf. Process. Manag.* **2009**, *45*, 427–437.
66. Chicco, D.; Jurman, G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics* **2020**, *21*, 6.
67. Platt, J.C. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*; Smola, A.J., Bartlett, P., Schölkopf, B., Schuurmans, D., Eds.; MIT Press: Cambridge, MA, USA, 1999; pp. 61–74.
68. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On calibration of modern neural networks. In Proceedings of the 34th International Conference on Machine Learning, Sydney, Australia, 6–11 August 2017; pp. 1321–1330.
69. Vickers, A.J.; Elkin, E.B. Decision curve analysis: A novel method for evaluating prediction models. *Med. Decis. Making* **2006**, *26*, 565–574.
70. Collins, G.S.; Reitsma, J.B.; Altman, D.G.; Moons, K.G.M. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement. *BMJ* **2015**, *350*, g7594.
71. Collins, G.S.; Moons, K.G.M.; Dhiman, P.; Riley, R.D.; Beam, A.L.; Van Calster, B.; Ghassemi, M.; Liu, X.; Reitsma, J.B.; van Smeden, M.; et al. TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* **2024**, *385*, e078378.